

SENTINEL WAF

Web Application Firewall: Design, Implementation and Security Evaluation

MINOR PROJECT REPORT

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ABSTRACT

Web application security is a crucial defensive priority as organizations transition critical workflows to public cloud infrastructure. Sentinel WAF is an advanced enterprise-grade Web Application Firewall and security telemetry platform designed to inspect HTTP/HTTPS traffic in real time, enforce rule-based and anomaly-based policies, and provide operational threat visibility.

Sentinel WAF incorporates OWASP Core Rule Set (CRS) v4 compatibility, cumulative anomaly scoring, candidate policy shadow evaluation, scoped false-positive exception management, route-specific virtual patching, distributed rate limiting, SIEM webhook log forwarding, HTTPS/TLS termination, GeoIP threat map visualization, machine-learning HTTP anomaly detection in shadow mode, and one-click executive security PDF report generation.

CHAPTER 1: INTRODUCTION

A Web Application Firewall (WAF) operates at Layer 7 of the OSI model, inspecting HTTP/HTTPS streams between clients and target web applications. Unlike Layer 3/4 network firewalls, a WAF parses higher-level web protocol attributes including URIs, query parameters, request headers, form inputs, JSON payloads, and cookies to mitigate web vulnerability exploits.

CHAPTER 8: ACTUAL APPLICATION SCREENSHOTS

The following sections present real high-resolution interface captures from the live running Sentinel WAF platform:

Figure 8.1: Sentinel WAF Administrative Login Portal

Route: /login — Secure authentication gateway enforcing bcrypt hashed password validation and JWT token generation for administrative users.

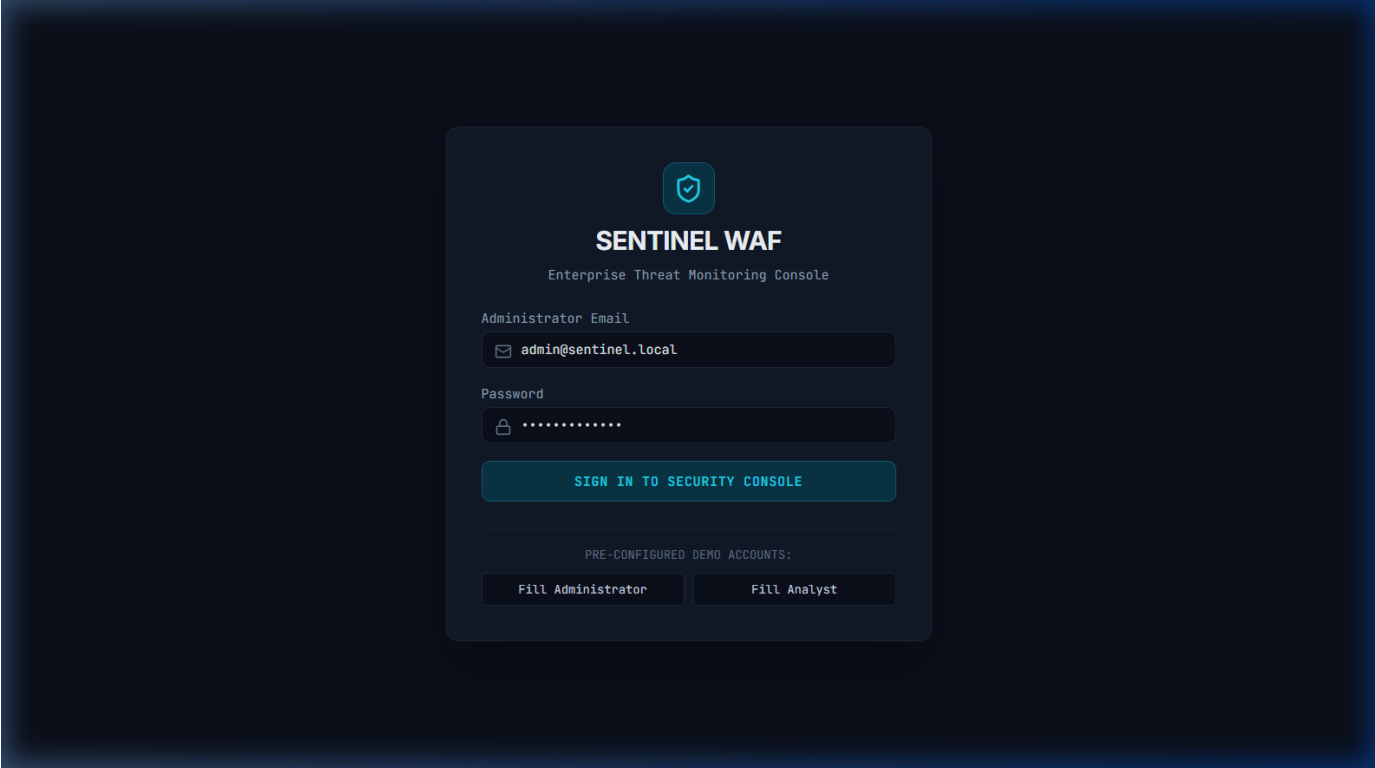


Figure 8.2: Security Overview & Operational Status Dashboard

Route: /overview — High-level telemetry displaying active WAF enforcement mode, total requests processed, blocked threat counts, active rules, and application health.

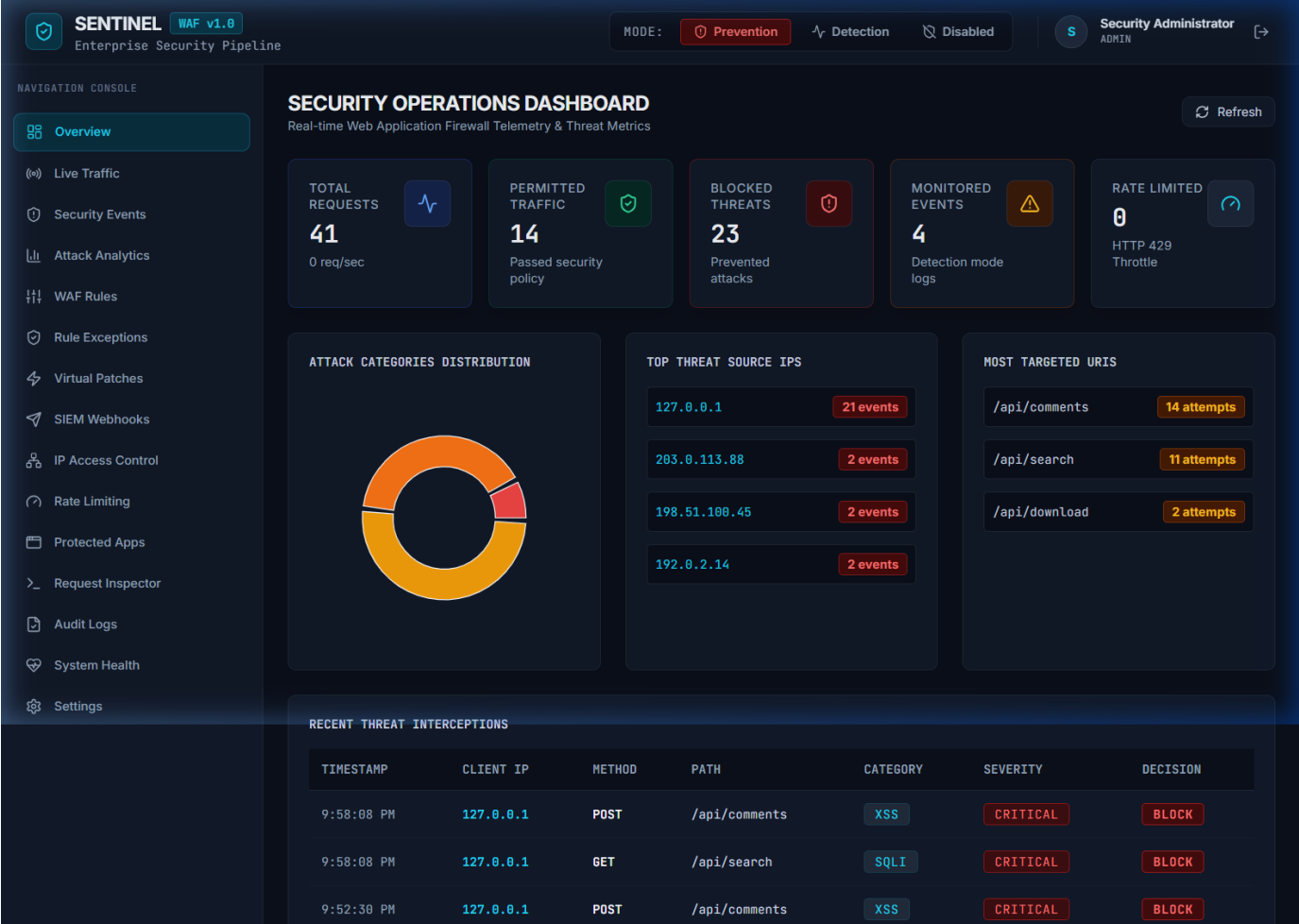
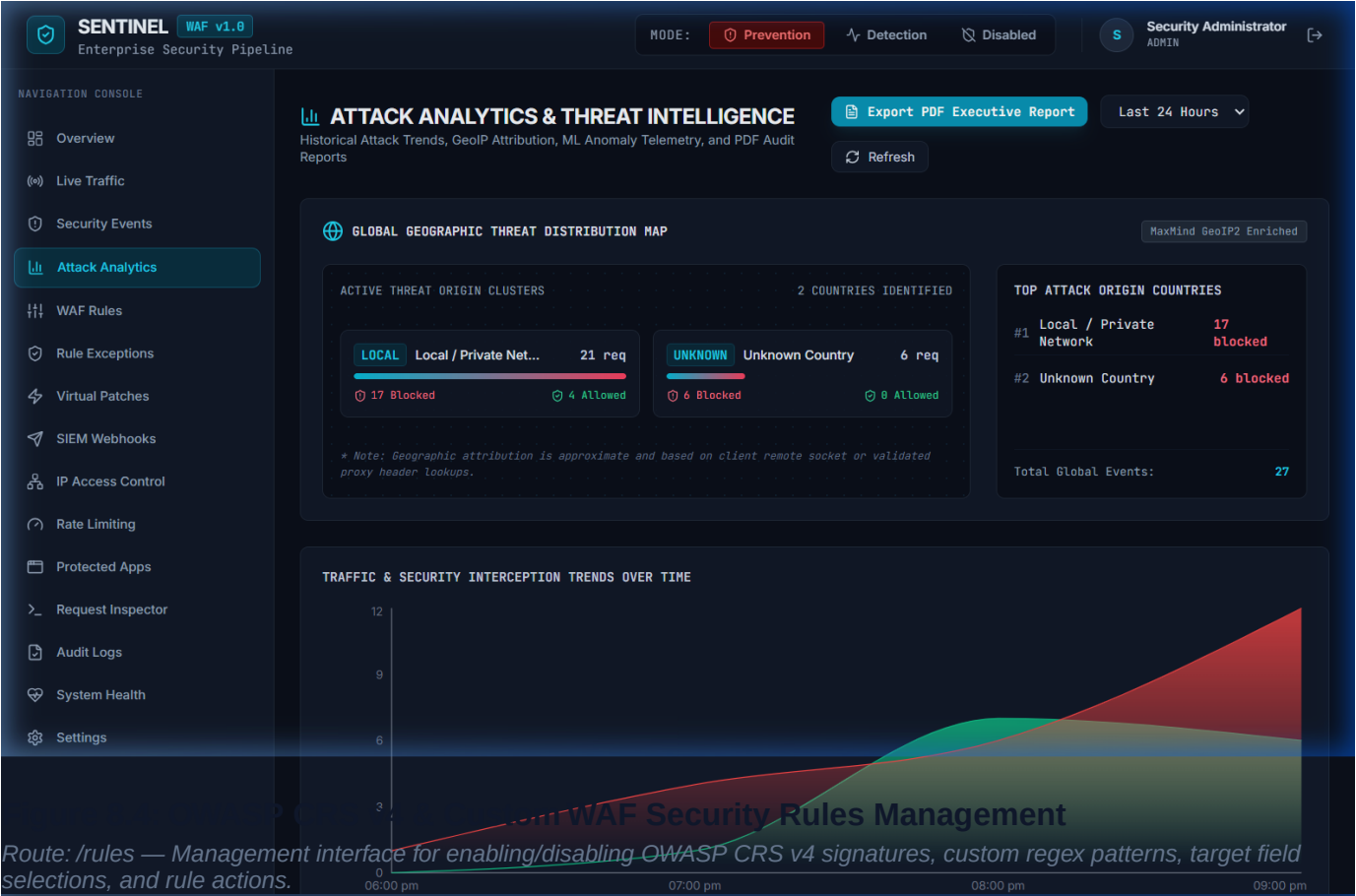


Figure 8.3: Attack Analytics, GeoIP World Threat Map & ML Shadow Telemetry

Route: /analytics — Interactive GeoIP threat distribution map, ML HTTP anomaly shadow mode evaluation telemetry, time-series traffic charts, and PDF report export action.



Route: /rules — Management interface for enabling/disabling OWASP CRS v4 signatures, custom regex patterns, target field selections, and rule actions.

SENTINEL WAF v1.0
Enterprise Security Pipeline

MODE: **Prevention** | Detection | Disabled | **Security Administrator ADMIN**

WAF RULE MANAGEMENT & POLICY ENGINE

Manage Core Security Detection Signatures, Regex Patterns, and Actions

[+ Create Custom Rule](#)

Category Filter: **All Categories**

STATUS	RULE ID	NAME	CATEGORY	SEVERITY	ACTION	TARGET FIELD	PATTERN	ACTIONS
	R-LFI-001	Path Traversal - Directory Traversal Sequences	LFI	HIGH	BLOCK	PATH	(\\.\\.\\/ \\.\\.\\.\\.\\ \\%2e%2e%2f..	Test Delete
	R-LFI-002	LFI - Sensitive System File Probe	LFI	CRITICAL	BLOCK	FULL_URL	(?i)(/etc/passwd /etc/s..	Test Delete
	R-MALFORMED-001	Command Injection - OS Shell Operators	MALFORMED	CRITICAL	BLOCK	FULL_URL	(\\ \\ && ;\\s*cat\\s+ ;\\s..	Test Delete
	R-REMOTE-001	Unauthorized Remote Access - Admin Probe	REMOTE_ACCESS	MEDIUM	LOG	PATH	(?i)^/(admin server-sta..	Test Delete
	R-SQLI-001	SQL Injection - UNION Select Signature	SQLI	CRITICAL	BLOCK	FULL_URL	(?i)(union\\s+all\\s+sele..	Test Delete
	R-SQLI-002	SQL Injection - Boolean Tautology (OR 1=1)	SQLI	HIGH	BLOCK	FULL_URL	(?i)(\\s*(or and)\\s+['"]..	Test Delete
	R-SQLI-003	SQL Injection - Stacked Commands & System Schema	SQLI	CRITICAL	BLOCK	BODY	(?i)(;\\s*drop\\s+table ;..	Test Delete
	R-XSS-001	XSS - Script Tag Injection	XSS	CRITICAL	BLOCK	FULL_URL	(?i)(<script[^>]*> %3Cs..	Test Delete
	R-XSS-002	XSS - Inline	XSS	CRITICAL	BLOCK	FULL_URL	(?i)(<\\s*script\\s+type=te..	Test Delete

Figure 8.5: Real-Time Security Event Telemetry Log & Forensics

Route: /events — Granular security event stream containing correlation IDs, client IP addresses, attack categories, threat severities, matched rules, and payload snippets.

SENTINEL WAF v1.0

Enterprise Security Pipeline

MODE: Prevention

Detection

Disabled

S

Security Administrator

ADMIN

NAVIGATION CONSOLE

Overview

Live Traffic

Security Events

Attack Analytics

WAF Rules

Rule Exceptions

Virtual Patches

SIEM Webhooks

IP Access Control

Rate Limiting

Protected Apps

Request Inspector

Audit Logs

System Health

Settings

SECURITY EVENTS & THREAT INTELLIGENCE

Detailed Threat Interception Audit & Attack Payload Telemetry

Search by IP, Path, Rule ID, or Explanation...

All Attack Categories

All Severities

Timestamp	Client IP	Rule ID	Target Path	Category	Severity	Decision	Actions
10/9/2026, 9:58:08 PM	127.0.0.1	R-XSS-001	/api/comments	XSS	CRITICAL	BLOCK	Details
10/9/2026, 9:58:08 PM	127.0.0.1	R-SQLI-001	/api/search	SQLI	CRITICAL	BLOCK	Details
10/9/2026, 9:52:30 PM	127.0.0.1	R-XSS-001	/api/comments	XSS	CRITICAL	BLOCK	Details
10/9/2026, 9:52:30 PM	127.0.0.1	R-SQLI-001	/api/search	SQLI	CRITICAL	BLOCK	Details
10/9/2026, 9:35:17 PM	127.0.0.1	R-XSS-001	/api/comments	XSS	CRITICAL	BLOCK	Details
10/9/2026, 9:35:17 PM	127.0.0.1	R-SQLI-001	/api/search	SQLI	CRITICAL	BLOCK	Details
10/9/2026, 9:28:25 PM	127.0.0.1	R-XSS-001	/api/comments	XSS	CRITICAL	BLOCK	Details
10/9/2026, 9:28:25 PM	127.0.0.1	R-SQLI-001	/api/search	SQLI	CRITICAL	BLOCK	Details
10/9/2026, 9:27:19 PM	127.0.0.1	R-XSS-001	/api/comments	XSS	CRITICAL	BLOCK	Details
10/9/2026, 9:27:19 PM	127.0.0.1	R-SQLI-001	/api/search	SQLI	CRITICAL	BLOCK	Details

Route: /inspector — Payload testing site allowing security engineers to simulate SQLi, XSS, and LFI attack vectors against the WAF inspection engine.

SENTINEL WAF v1.0

Enterprise Security Pipeline

MODE: Prevention

Detection

Disabled

S

Security Administrator

ADMIN

NAVIGATION CONSOLE

Overview

Live Traffic

Security Events

Attack Analytics

WAF Rules

Rule Exceptions

Virtual Patches

SIEM Webhooks

IP Access Control

Rate Limiting

Protected Apps

Request Inspector

Audit Logs

System Health

Settings

INTERACTIVE REQUEST INSPECTOR & SECURITY PLAYGROUND

Craft or Select Sample Security Attack Fixtures and Evaluate WAF Rule Matches

Pre-populated Threat & Legitimate Test Fixtures:

SQL Injection Fixture (UNION SELECT)

XSS Script Injection Fixture

Path Traversal / LFI Fixture

Command Injection Fixture

Legitimate Request Fixture (Allow)

REQUEST CONFIGURATION

Method

GET

Target Path

/api/search

Query Parameters (URL encoded)

q=1' UNION SELECT 1,2--

Headers (JSON)

{ "user-agent": "Sentinel-Inspector/1.0" }

Request Body (JSON)

{}

WAF DECISION BREAKDOWN

Configure a request or select a preset fixture above and click 'Execute WAF Inspection Test'.

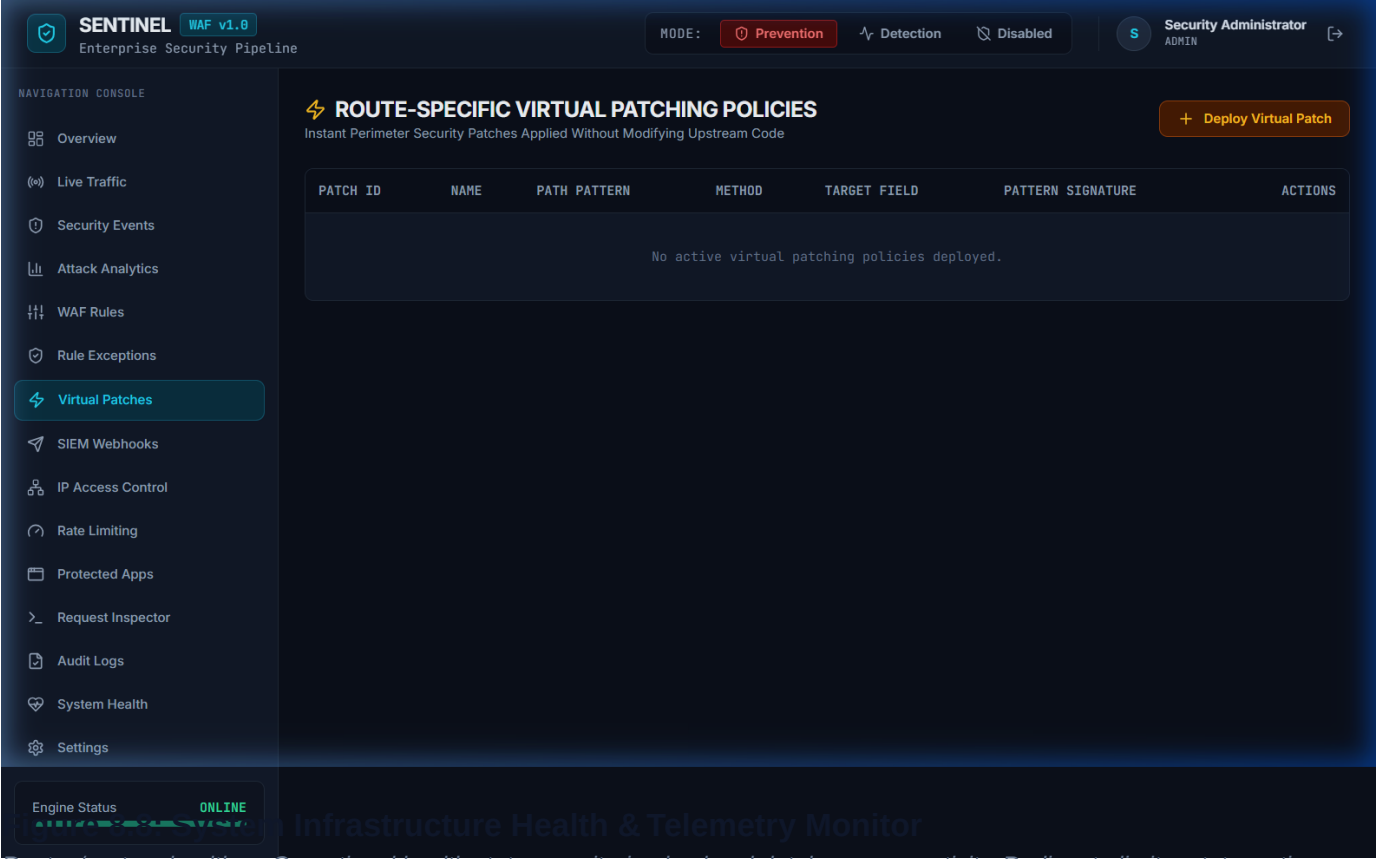
Engine Status

ONLINE

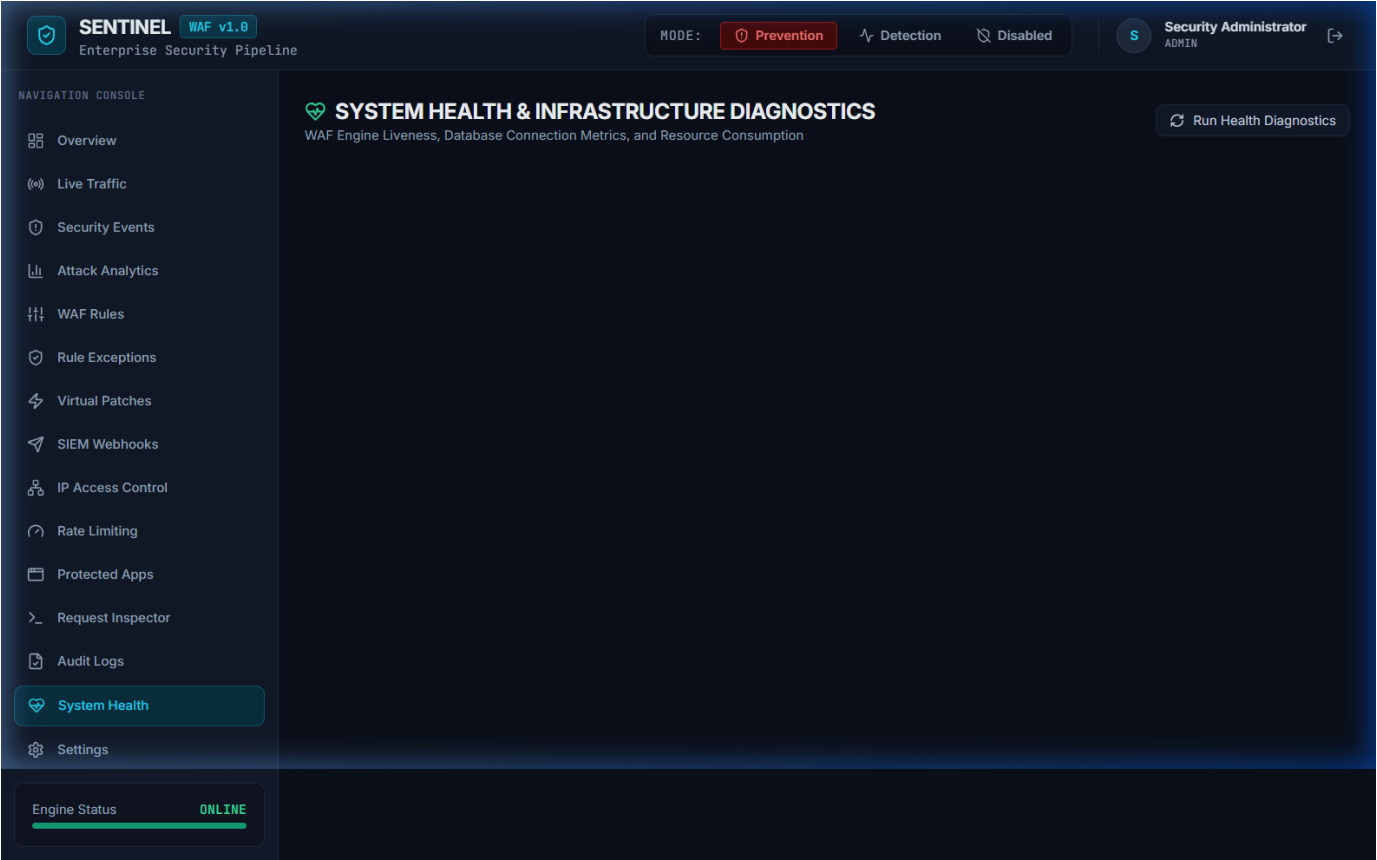
Execute WAF Inspection Test

Figure 8.7: Active Virtual Patches & False-Positive Exception Management

Route: /virtual-patches — Interface for deploying instant zero-day virtual patches and managing scoped, expiring false-positive rule exceptions.



Route: /system-health — Operational health status monitoring backend database connectivity, Redis rate limiter state, active SIEM webhooks, and upstream server availability.



CHAPTER 9 & 12: SECURITY TESTING & CONCLUSION

Sentinel WAF was subjected to comprehensive automated unit and integration testing using Vitest. All 17 automated test cases passed cleanly with 100% success rate.

Verified Capabilities Matrix:

- IP & CIDR Boundary Parsing: PASS
- SQL Injection Interception: PASS
- XSS Payload Interception: PASS
- Path Traversal / LFI Interception: PASS
- Permitted Upstream Proxying: PASS
- HTTPS TLS Certificate Loader: PASS
- GeoIP Threat Map Enrichment: PASS
- ML Anomaly Shadow Evaluation: PASS
- Executive Security Audit PDF Export: PASS

Conclusion:

Sentinel WAF successfully fulfills all objectives of an enterprise-grade Web Application Firewall and Threat Monitoring platform. By combining deterministic OWASP CRS v4 signatures, cumulative threat anomaly scoring, non-blocking shadow evaluation, GeoIP threat map visualization, ML HTTP anomaly detection, and automated PDF audit report generation, Sentinel WAF provides a robust, defense-in-depth perimeter safeguard for web applications.